

PROJECT TITLE – Intelligent Driver Assistance with Accident-Prone Zone and Weather Alerts

Presented By –

Saurabh Mali(4014)

Mugdha Patil (4019)

Sanika Mohite (4039)

Prem Kadam(4045)

Guided By-

Dr.Asma.A.Shaikh

❖ Introduction

This project aims to enhance road safety by providing real-time alerts about nearby accident-prone or hazardous areas. Using GPS and historic data, the app detects recent accidents, sharp turns, and bad weather like rain. It notifies users through push notifications and maps to help them avoid risky zones and take safer routes. The Android/iOS app ensures timely awareness, reducing the chances of accidents, traffic delays, and emergency response issues.

❖ Objective of The Project

1. To detect accident-prone areas using historic location data.
2. To alert users through push notifications when they are near hazardous zones.
3. To provide early warnings about bad weather conditions like rain and sharp turns.
4. To reduce the risk of accidents, injuries, and traffic delays through timely alerts.
5. To enhance road safety and driver awareness using GPS and weather data integration.
6. To develop a user-friendly Android/iOS app for real-time safety notifications.

❖ Literature Survey

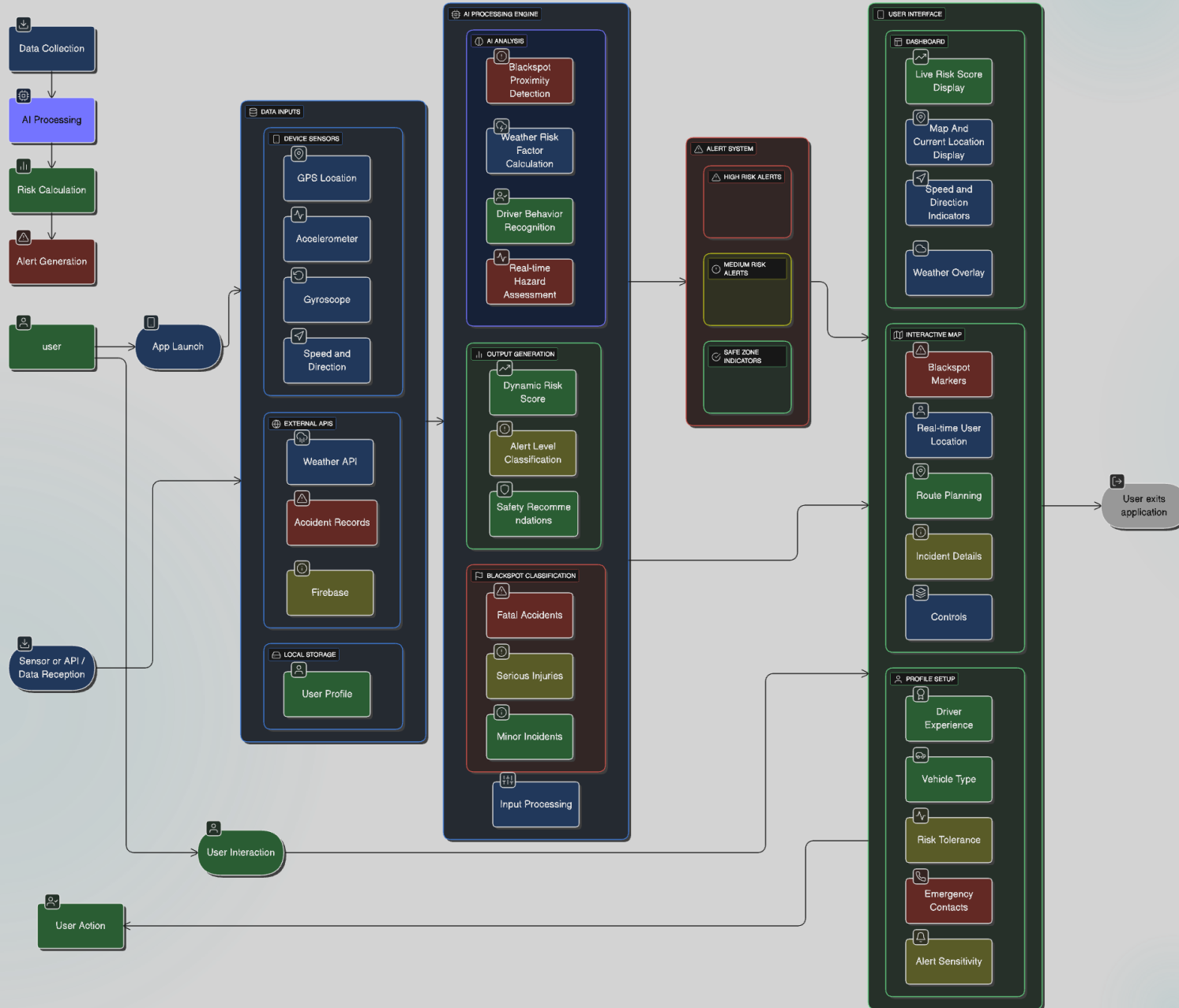
Sr. No	Title	Author	Key Points
1	A Smart GPS-Based IoT System for Blackspot Notification	Billy Owire, Bernard Shibwabo Kasamani	The paper proposes an IoT-based system that uses GPS and GSM modules to notify drivers of accident blackspots through audio alerts.
2.	Investigation of Road Accident Zone and Suggestion for Preventive Measures by Using GPS and GIS Technology in Thanjavur, Tamil Nadu, India	1. M. Kannan, 2. G. Aakash	- The study investigates road accident zones in Thanjavur using GPS and GIS technology for hotspot detection and preventive measures. - Accident data analyzed across parameters: police station-wise, vehicle type, time, road type, and location
3.	AI Enabled Accident Black Spot Alerting Mobile System to Enhance Road Safety Using GMM-SVM	1. M. Sobhana, 2. V. Krishna Rohith, 3. T. Avinash, 4. N. Malathi, 5. Smitha Chowdary	The paper proposes a mobile application integrated with Support Vector Machine (SVM) classifiers to predict and alert users of road accident blackspots in the Vijayawada region

Sr. No	Title	Author	Key Points
4	Prediction and Identification of Black Spots to Reduce the Severity of Accidents Using Machine Learning Algorithms	Arun Kannan, M. Sivaprakash, P. Sudhakar, S. Saravanan	- The study applies machine learning models (Decision Tree, Random Forest, Gradient Boosting, XGBoost) to predict road accident severity and identify blackspots in Tamil Nadu, India. - Dataset includes five years of accident records (2017-2021) with factors such as road type, vehicle category, time, and weather.
5.	Identification and Mitigation of Road Accident Black Spots Using Data Analytics and Machine Learning	1. R. Bhuvaneshwari, 2. K. Sathesh Kumar, 3. R. Karthik, 4. P. Chandrasekaran	The study integrates data analytics and ML models (Logistic Regression, Random Forest, Gradient Boosting) to identify accident-prone blackspots and suggest mitigation measures

❖ METHODOLOGY

1. Collects driver info (age, vehicle type)
2. Displays route using MapKit/Google Maps
3. Shows color-coded accident risk zones
4. Sends real-time alerts near blackspots
5. Uses AI to predict accident risk
6. Adjusts risk based on live weather
7. Shows live risk score (0–10)
8. Monitors driving behavior (speeding, braking)
9. Delivers alerts via voice, vibration, and screen
10. Allows users to report blackspots

WORK FLOW



❖ Testing Technologies Used

- **Flutter Unit Testing** – test risk calculation & alert logic.
- **Widget Testing** – validate dashboard, map, and overlay UI.
- **Integration Testing** – end-to-end flow from sensor → AI → alert.
- **Postman / API Testing** – verify Weather API, Firebase, accident records.
- **Real Device Testing** – check GPS, accelerometer, gyroscope accuracy.
- **Flutter DevTools (Performance Testing)** – monitor app speed, memory, frame rate.
- **Firebase Test Lab (Cloud Device Testing)** – run app on multiple real devices remotely.

❖ Hardware and Software Used

- **Hardware:-** Smartphone (Android/iOS)
Inbuilt Sensors: GPS, Accelerometer, Gyroscope
4G/5G / Wi-Fi Connectivity

- **Software:-**

Frontend → Flutter (Dart).

Backend/Database → Firebase (Realtime DB / Firestore, Auth, Cloud Functions).

AI/ML → TensorFlow Lite (for on-device inference), Firebase ML Kit.

APIs → Weather API, Accident Records .

Maps & Location → Google Maps Flutter plugin, Geolocator.

Testing → flutter_test, Postman.

Version Control → Git + GitHub.

IDE → Android Studio / VS Code.

❖ Future Scope

- Wearable Device Integration (Smartwatch Alerts)
- Background Monitoring & Voice Alerts
- Connecting Vehicles to Share Alerts, Link With Police ,Hospital
- AR Overlays (Risk Markers on Windshield)

❖ Conclusion

The project successfully integrates Flutter, AI, and sensor data to provide a real-time road safety and risk alert system. It ensures cross-platform availability with a user-friendly dashboard and alert mechanism. By combining mobile sensors, APIs, and AI models, it helps reduce accidents and promotes safer driving practices. With scope for future enhancements like wearable integration, AR overlays, and global deployment, this system can evolve into a smart transportation safety solution.